

Cardia Atlas

Cardiovascular Risk Visualization & Prediction · Track A

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Project summary. Cardia Atlas is a browser application, deployed publicly and runnable locally, for estimating overall coronary artery disease (CAD) and LAD, LCX, and RCA stenosis probabilities from clinical, examination, ECG, laboratory, and echocardiographic features. It integrates four trained model pipelines, patient-specific SHAP explanations, and an interactive original 3D coronary schematic.

The application is an educational decision-support prototype. It is not a diagnosis, a validated clinical risk calculator, or a substitute for clinical assessment or diagnostic imaging. Model probabilities are not measured stenosis percentages or lesion locations.

What is implemented

A responsive web interface accepts known clinical inputs and permits blank fields; fitted imputers handle unavailable entries. The FastAPI API returns four model probabilities and approximate SHAP contributions. The Three.js view supports rotation, zoom, and vessel selection. Performance and limitations are visible in the application and in this report.

Key data and validation facts

303 records	54 predictors	4 predicted outcomes	2× outer CV repeats
UCI primary sheet	after target + constant exclusions	CAD, LAD, LCX, RCA	5-fold outer validation

Access and quick start

Public app: cardia-atlas.onrender.com · **Source:** [GitHub repository](#). Run locally with Python 3.14+: `bash scripts/run_local.sh`; this installs the pinned requirements and starts the app at `127.0.0.1:8000`. First launch requires package-index access.

Load synthetic demo values, choose Generate analysis, edit values and generate again as needed, then rotate or zoom the 3D view and select an artery to inspect its prediction and explanation. Clear all resets the form and results. The visible explanation lists the 10 largest absolute SHAP contributions; full feature contributions are used for the additivity check.

Data, targets & leakage controls

The primary sheet of the official UCI Extension of Z-Alizadeh Sani workbook contains **303 records and 59 columns**. UCI describes CAD as at least 50% diameter narrowing; CAD is associated with one or more stenotic LAD, LCX, or RCA labels. The positive targets here are Cath=CAD and each vessel=Stenotic.

Observed source target balance

Outcome	Positive	Negative	Source encoding
CAD	216	87	CAD / Normal
LAD	177	126	Stenotic / Normal
LCX	119	184	Stenotic / Normal
RCA	114	189	Stenotic / Normal

Audit results: zero missing values, zero exact duplicate rows, no row identifier in the primary sheet. There are 55 non-target fields before removing one invariant field; the resulting schema contains 54 predictors. The four target fields Cath, LAD, LCX, and RCA are explicitly excluded from all four predictor matrices. No target values enter preprocessing, model fitting, or SHAP backgrounds.

Leakage and preprocessing decisions

The 303 observed `Exertional CP` values are all N, so this constant input was removed. The source value `Fmale` was normalized to `Female`; no row was removed or relabeled. The source CAD label was retained as supplied: 302/303 Cath labels agree with the logical OR of vessel labels, leaving one internal inconsistency documented in the audit. Clinical, ECG, laboratory, and echo fields described by the brief were otherwise kept, including Region RWMA as an echocardiographic input, not a lesion coordinate.

For each fold, median/mode imputers, one-hot category encoding, and numeric scaling are fit only on the training partition. The source has no missing values, but the local form accepts blanks and uses these training-fitted imputers. No resampling or feature selection is used. Numeric UI values are limited to observed source ranges; UCI does not specify units for many variables, so units are not guessed.

Dataset citation: Alizadehsani, R., Roshanzamir, M., & Sani, Z. (2013). *extension of Z-Alizadeh sani dataset*. UCI Machine Learning Repository. DOI 10.24432/C5461K; CC BY 4.0.

Model selection & evaluation

Three fixed pipeline families were compared: regularized Logistic Regression (balanced class weights), Extra Trees, and Random Forest. Nested repeated stratified 5-fold outer validation (two repeats) estimates performance. In each outer training partition, 3-fold ROC-AUC selects among the families; the held-out outer fold provides the estimate. The 0.50 threshold is fixed, not optimized. The final pipeline family is chosen separately via 5-fold ROC-AUC on all records and then refit on all records. This final refit has no independent holdout.

Outer-fold performance (mean $\pm$ standard deviation)

Outcome	Final refit family	Acc.	Prec.	Sens.	Spec.	F1	ROC AUC	PR AUC	Brier	ECE
CAD	ExtraTrees	0.86 $\pm$ 0.05	0.92 $\pm$ 0.04	0.88 $\pm$ 0.06	0.80 $\pm$ 0.10	0.90 $\pm$ 0.04	0.92 $\pm$ 0.03	0.97 $\pm$ 0.01	0.11 $\pm$ 0.02	0.11 $\pm$ 0.05
LAD	RandomForest	0.77 $\pm$ 0.06	0.80 $\pm$ 0.07	0.83 $\pm$ 0.06	0.70 $\pm$ 0.13	0.81 $\pm$ 0.04	0.84 $\pm$ 0.05	0.88 $\pm$ 0.04	0.17 $\pm$ 0.02	0.12 $\pm$ 0.05
LCX	RandomForest	0.70 $\pm$ 0.07	0.63 $\pm$ 0.11	0.63 $\pm$ 0.10	0.75 $\pm$ 0.10	0.63 $\pm$ 0.08	0.75 $\pm$ 0.08	0.63 $\pm$ 0.08	0.21 $\pm$ 0.03	0.11 $\pm$ 0.04
RCA	Logistic regression	0.65 $\pm$ 0.05	0.54 $\pm$ 0.06	0.55 $\pm$ 0.11	0.71 $\pm$ 0.07	0.54 $\pm$ 0.07	0.72 $\pm$ 0.06	0.60 $\pm$ 0.07	0.21 $\pm$ 0.02	0.12 $\pm$ 0.04

Outer-fold metrics estimate the inner-CV model-selection procedure: each outer fold selects a family within its training data before the fitted pipeline is evaluated on held-out data. The "final refit family" column separately shows the family chosen by 5-fold ROC-AUC on all 303 records and refit on all records; the outer-fold scores do not evaluate that particular full-data refit. The threshold was fixed at 0.50. Sensitivity = recall; PR-AUC = average precision; ECE = 5-bin expected calibration error. Brier scores and ECE summarize probability reliability but do not validate calibration. Pooled held-out confusion counts (TN/FP/FN/TP, two out-of-fold repeats per record): CAD 140/34/52/380; LAD 175/77/62/292; LCX 277/91/88/150; RCA 269/109/102/126.

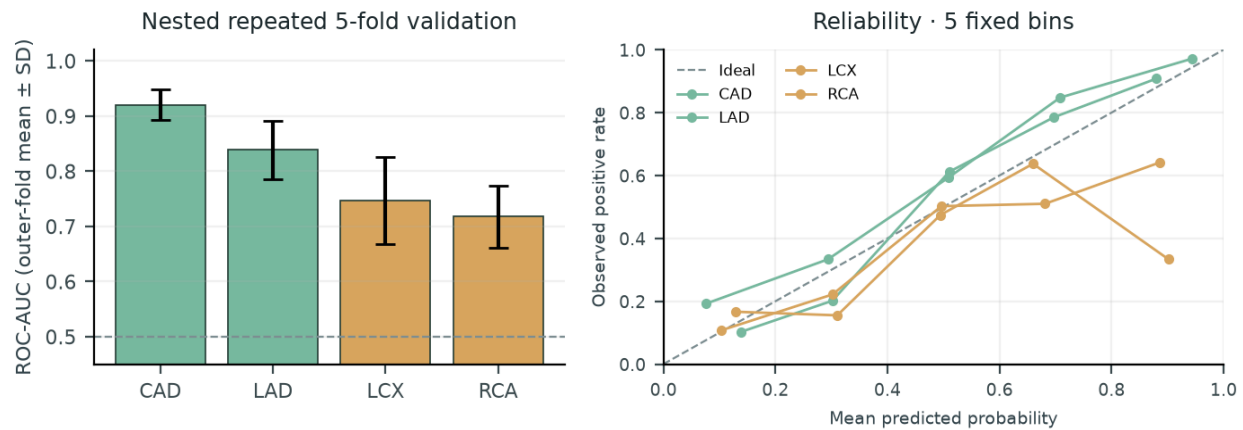


Figure 1. Left: ROC-AUC outer-fold means and SD from saved nested-CV results; error bars are fold-to-fold SD, not confidence intervals. Right: five-bin reliability from pooled outer-fold predictions; bin counts and small cohort make this a descriptive check, not calibration validation.

CAD performed best in these internal estimates. Vessel-specific results are weaker, especially LCX/RCA, with notable fold variability. The sample is small and targets imbalanced; model family selection can still create optimism despite the nested outer design. No external cohort was available.

Patient explanations & 3D visualization

After a user selects known clinical fields, the app API returns four positive-class probabilities and one explanation per target. The selected family in the final artifact is CAD Extra Trees, LAD Random Forest, LCX Random Forest, and RCA Logistic Regression. It was selected from all-data 5-fold ROC-AUC comparison and refit on all 303 records.

SHAP explains the actual fitted preprocessing-plus-estimator path: a fixed background of 24 source rows, six permutation cycles, and one-hot contributions summed back to their source feature. The displayed baseline and contributions use positive-class probability units. The browser lists the 10 largest absolute contributions; the additivity check uses the full 54-feature attribution before the chart is truncated. Blank values are identified as imputed. Finite permutations can vary, and correlated clinical variables complicate attribution. Contributions describe model associations, not causal effects.

Interactive schematic view

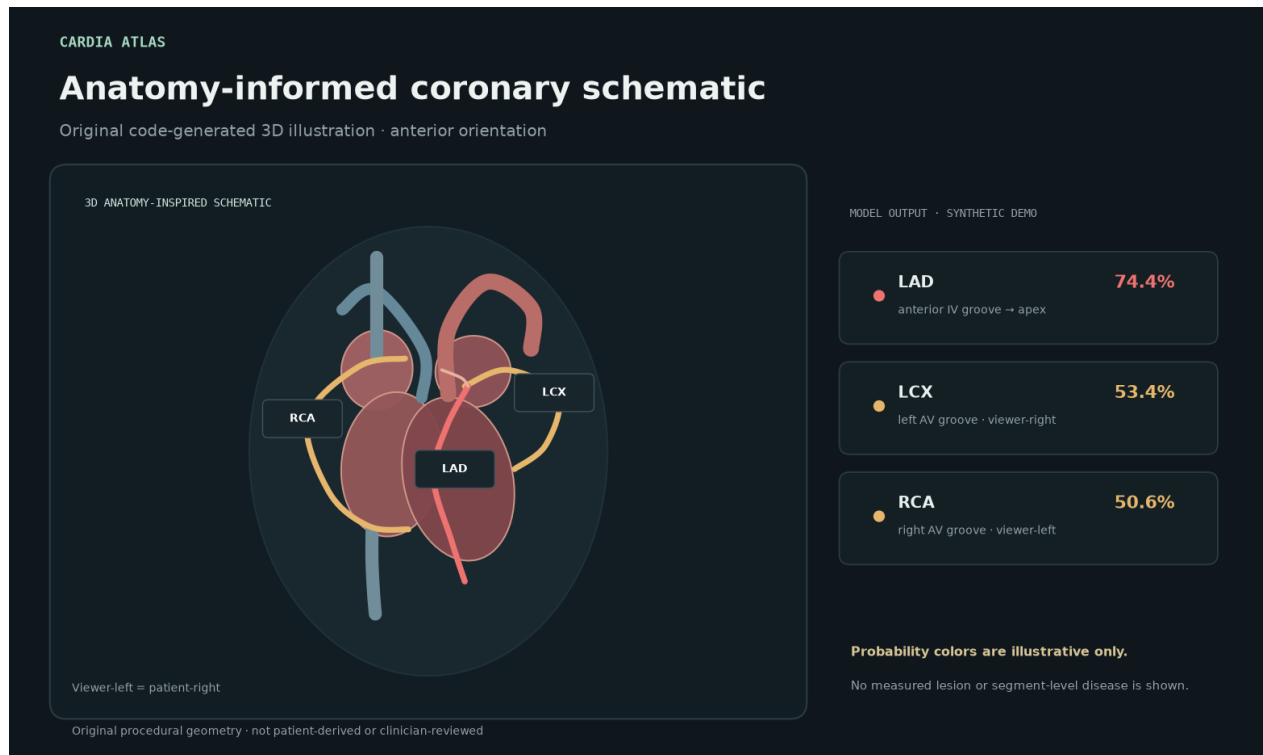


Figure 2. Original schematic diagram derived from the application geometry and synthetic median/mode demo inference. The 3D chamber and artery paths are original code-generated geometry, not a patient-derived or clinician-reviewed anatomical mesh. Vessel colors show model probabilities only.

The app uses an original, code-generated 3D chamber-and-great-vessel illustration with schematic coronary paths. In the anterior view, LAD follows the anterior interventricular groove toward the apex; LCX follows the left atrioventricular groove (patient-left/viewer-right); RCA follows the right atrioventricular groove (patient-right/viewer-left). These broad courses were checked against NHLBI and NCBI anatomy references. The model is not patient-derived or clinician-reviewed and does not claim segment-level accuracy or anatomical variants. Drag rotates; wheel or pinch zooms; artery markers, labels, and result cards select a vessel and its explanation. Continuous color bands are illustrative: [0,33%), [33,67%), and [67,100%]. The drawing does not place a measured stenosis on anatomy.

The form groups demographic, history/examination, ECG, and laboratory/echo features, displays source-observed numeric ranges, offers source-coded categorical choices, and accepts missing fields. It labels the feature-wise median/mode synthetic demo and marks edited profiles. The UI includes a visible educational safety disclaimer, model performance, and limitations.

Architecture, verification & limits

The browser application is plain JavaScript with bundled Three.js 0.180.0. The same FastAPI service is used locally and at the verified public Render URL; it validates the prediction contract and returns predictions and SHAP values. Four scikit-learn pipelines serialize preprocessing with each estimator. A health endpoint checks model presence; schema and performance endpoints serve the exact fit-time feature definition and saved experiment JSON. The original schematic is procedurally generated; no external anatomical mesh, image, paid service, or API key is used.

Checks completed

Automated unit tests: 8 passed in the repository environment and again from the unpacked submission archive in a newly created environment with the pinned requirements; the JavaScript continuous-band boundary checks also passed. A clean package install and inference check passed without model retraining. Verified source dimensions, class balance, missingness, duplicates, CAD/vessel consistency, median/mode demo fields, target-leakage exclusions, 54-field schema, all four saved models, and serialization/reload prediction equivalence. Integrated local API: health/schema/performance endpoints, valid synthetic demo predictions, all-fields-missing imputation, invalid-category HTTP 422, and friendly missing-artifact HTTP 503. SHAP: positive-class output, feature mapping, and baseline-plus-contributions additivity checked. Browser: Three.js WebGL canvas rendered; changing a field and submitting changed all four probabilities and color bins; selecting each vessel updated its explanation; drag rotation, wheel zoom, and reset were checked. The compact chart shows ten of 54 features; additivity is checked on the full attribution.

Not verified

A local Docker build/start was not run because the environment had no container engine; Render successfully built and started the deployed service. Cross-browser compatibility, a full screen-reader/accessibility audit, GPU-free speed on other devices, clinical review, clinical probability calibration validation or recalibration, external/temporal evaluation, and a recorded YouTube video were not completed. A 390 px layout and keyboard focus on core controls were smoke-checked on one browser/device only. Descriptive Brier, five-bin ECE, and reliability checks were computed on outer-fold predictions. Figure 2 is a code-derived schematic, not a screenshot or independent anatomical validation.

Important limitations

These 303 source records are one small cohort and do not demonstrate performance in other populations or current clinical settings. No independent validation, prospective study, calibration study, fairness analysis, or clinical threshold validation was performed. CAD class balance differs substantially from the vessel targets. The interface blocks values outside the observed source range, but values inside the range can still be invalid or clinically implausible. The schema provides no source units for many inputs. All performance claims are research summaries, not evidence of safety or clinical utility.

Source and asset attribution

Dataset: Alizadehsani R, Roshanzamir M, Sani Z. (2013). [UCI Machine Learning Repository](#), DOI [10.24432/C5461K](#), CC BY 4.0. Coronary course references: [NHLBI blood flow](#); [NCBI Bookshelf LAD anatomy](#); and [Coronary Artery Anomalies review](#). The heart, chambers, great vessels, and coronary paths are original procedural Three.js geometry; no external anatomy mesh, image, or icon was imported. [Three.js 0.180.0](#), MIT License; original library headers are retained. SHAP: [PermutationExplainer documentation](#). OpenAI Codex assisted implementation, model workflow, interface, checks and documentation; this is disclosed in README and should be entered in Devpost Built With.

Track A source: participant-provided four-page Track A brief. Official event rules, eligibility, deadline, and submission notes are linked in the project README. Report length is limited to six pages; the required demonstration video is 3–10 minutes and is not yet recorded. The timed demo script is included separately.